

Master's Degree in Quantum Technologies

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Finite temperature variational quantum machine learning

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Abstract

This work develops a variational method to approximate the thermal state of a quantum many-body system at finite temperature, avoiding the exponential cost of exact diagonalization. Instead of working with the thermal density matrix directly, we purify the system by introducing an ancillary set of spins and seek a pure state of the combined system. The von Neumann entropy is replaced by the second Rényi entropy S_2 , which, once the ancilla is introduced, becomes the entanglement entropy between system and ancilla and can be estimated via sampling. Minimizing the resulting Rényi free energy defines the maximal Rényi ensemble (MRE), which we approximate variationally using Neural Quantum States, parameterized by autoregressive neural networks and implemented within the NetKet framework.

We apply this method to the one-dimensional transverse-field Ising model. The variational results are validated against exact diagonalization for small systems, correctly reproducing the analytical MRE solution. We then study the scaling of the method with system size, confirming numerically that local observables of the MRE converge to those of the Gibbs ensemble in the thermodynamic limit, provided the two ensembles are compared at the same mean energy rather than at the same temperature. We further analyze the role of the ancilla size, showing that an ancilla smaller than the physical system already reproduces the relevant thermal properties at low temperatures, substantially reducing the computational cost. Finally, we characterize the main limitation of the method: the statistical estimation of S_2 via the swap trick suffers from noise that grows exponentially with the entropy, which restricts the temperature range that can be reliably accessed despite the use of an improved λ -integral estimator.

Resumen

Este trabajo desarrolla un método variacional para aproximar el estado térmico de un sistema cuántico de muchos cuerpos a temperatura finita, evitando el coste exponencial de la diagonalización exacta. En lugar de trabajar directamente con la matriz densidad térmica, purificamos el sistema introduciendo un conjunto auxiliar de espines (ancilla) y buscamos un estado puro del sistema conjunto. La entropía de von Neumann se sustituye por la segunda entropía de Rényi S_2 , que, una vez introducida la ancilla, se convierte en la entropía de entrelazamiento entre sistema y ancilla y puede estimarse mediante muestreo. Minimizar la energía libre de Rényi resultante define el ensemble maximal de Rényi (MRE), que aproximamos variacionalmente mediante Estados Cuánticos Neuronales, parametrizados por redes neuronales autorregresivas e implementados dentro del marco de NetKet.

Aplicamos este método al modelo de Ising unidimensional con campo transversal. Los resultados variacionales se validan frente a la diagonalización exacta para sistemas pequeños, reproduciendo correctamente la solución analítica del MRE. A continuación estudiamos el escalado del método con el tamaño del sistema, confirmando numéricamente que los observables locales del MRE convergen a los del ensemble de Gibbs en el límite termodinámico, siempre que ambos ensembles se comparen a la misma energía media y no a la misma temperatura. Analizamos además el papel del tamaño de la ancilla, mostrando que una ancilla más pequeña que el sistema físico ya reproduce las propiedades térmicas relevantes a bajas temperaturas, reduciendo sustancialmente el coste computacional. Por último, caracterizamos la principal limitación del método: la estimación estadística de S_2 mediante el swap trick sufre un ruido que crece exponencialmente con la entropía, lo que restringe el rango de temperaturas al que se puede acceder de forma fiable, a pesar del uso de un estimador mejorado basado en la integral en λ .

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1. Introduction

The quantum many-body problem refers to the challenge of describing the collective behavior of a large number of interacting quantum particles. It underlies a wide range of physical phenomena, from phase transitions and magnetism to superconductivity, and encompasses several distinct sub-problems: finding ground states, simulating real-time dynamics, and computing thermal properties, among others. This work focuses on the last of these: how to represent and characterize the state of a quantum system at finite temperature.

At finite temperature, a quantum system is not described by a single wave function but by a statistical mixture, encoded in the thermal density matrix $\hat{\rho}_{\text{thermal}}$, which depends on the Hamiltonian $\hat{H}$ and the inverse temperature β .

$$\hat{\rho}_{\text{thermal}} = \frac{e^{-\beta\hat{H}}}{\mathcal{Z}}, \quad \mathcal{Z} = \text{tr} \left(e^{-\beta\hat{H}} \right). \quad (1.1)$$

Working directly with $\hat{\rho}_{\text{thermal}}$ is difficult: density matrices are positive, Hermitian operators subject to normalization and positivity constraints that are awkward to enforce variationally, and their dimension grows as 2^N for N spin-1/2 particles, making exact diagonalization intractable beyond a few tens of sites.

A standard strategy to circumvent these difficulties is purification: instead of working with a mixed state ρ_S of the physical system S , one introduces an auxiliary system A and seeks a pure state $|\Psi\rangle_{SA}$ such that ρ_S is recovered by tracing out the ancilla, $\rho_S = \text{tr}_A(|\Psi\rangle\langle\Psi|)$. This trades the constraints of mixed-state optimization for the more tractable problem of representing a pure state in a larger Hilbert space. In this work, the purification is implemented precisely by appending an ancillary set of spins to the physical system and later tracing it out.

Representing this enlarged pure state still requires a method that avoids the exponential growth of the Hilbert space. Several variational approaches address this challenge, most notably tensor networks (TN), which exploit the limited entanglement of physical states to compress the wave function, and Neural Quantum States (NQS), which instead parameterize the wave function amplitudes using a neural network with a number of parameters that scales only polynomially with system size. This work adopts the NQS approach.

A further obstacle remains even after purification: the natural measure of mixedness, the von Neumann entropy $S_{\text{vN}} = -\text{tr}(\rho \ln \rho)$, cannot be written as the expectation value

of a simple observable, and computing it exactly requires diagonalizing ρ and knowing its entire spectrum, which is again intractable for large systems. To resolve this, we replace S_{vN} with the second Rényi entropy S_2 , which has the crucial advantage that, once the ancilla is introduced, it can be estimated via Monte Carlo sampling without ever diagonalizing the density matrix. Minimizing the resulting Rényi free energy defines the maximal Rényi ensemble (MRE), which we use throughout this work as a tractable proxy for the thermal state.

With this framework in place, the present work pursues two main objectives: first, to verify that the variational optimization of the purified state correctly converges to the MRE as the ancilla is introduced, and second, to characterize the similarities and differences between the resulting Rényi ensemble and the canonical Gibbs ensemble.

2. Framework and methodology

2.1 Model system

We study the one-dimensional transverse-field Ising model with nearest-neighbor interactions and periodic boundary conditions, described by the Hamiltonian

$$\hat{H} = h_x \sum_{i=1}^N \hat{\sigma}_i^x + J_{zz} \sum_{i=1}^N \hat{\sigma}_i^z \hat{\sigma}_{i+1}^z, \quad (2.1)$$

where $\hat{\sigma}_i^x$ and $\hat{\sigma}_i^z$ are the Pauli matrices at site i , h_x is the strength of the transverse magnetic field, J_{zz} is the nearest-neighbor coupling in the z direction, and N is the number of spins in the chain, with $\hat{\sigma}_{N+1} \equiv \hat{\sigma}_1$ due to periodic boundary conditions. This model is a paradigmatic testbed in quantum many-body physics: its integrability via the Jordan–Wigner transformation provides access to analytical results, which makes it particularly well suited for benchmarking the variational method developed in this work.

To benchmark the variational method against exact results, we focus on two local observables that are also experimentally accessible in the laboratory: the transverse magnetization $\langle \sigma^x \rangle$ and the nearest-neighbor correlation functions

$$\Gamma^{x,x} = \frac{1}{N} \sum_{i=1}^N \langle \hat{\sigma}_i^x \hat{\sigma}_{i+1}^x \rangle, \quad \Gamma^{z,z} = \frac{1}{N} \sum_{i=1}^N \langle \hat{\sigma}_i^z \hat{\sigma}_{i+1}^z \rangle. \quad (2.2)$$

Both quantities are local in the sense relevant to the thermodynamic-limit arguments discussed below, and their experimental accessibility makes them natural targets for assessing whether the method correctly reproduces thermal expectation values.

2.1.1 Density of states

The integrability of the one-dimensional quantum Ising model via the Jordan–Wigner (JW) transformation [1] allows analytical approximations to the density of states $g(E)$ to be derived. This quantity will be needed later for the analytical evaluation of the energy probability distribution $p(E)$ of the two ensembles (Section 3.2.3), and we introduce it here since it is a property of the model itself, independent of the variational method.

The first is the saddle-point approximation, which follows from the inverse Laplace trans-

form representation of $g(E)$,

$$g(E) = \frac{1}{2\pi i} \int Z(\beta) e^{\beta E} d\beta, \quad (2.3)$$

evaluated at the saddle point β^* defined by $\langle E \rangle_{\beta^*} = E$. This yields

$$\ln g(E) \approx \beta^* EN + \ln Z(\beta^*), \quad (2.4)$$

which is exact in the thermodynamic limit $N \rightarrow \infty$ but smooths over any discrete structure of the spectrum at finite N .

The second approximation is the multi-Gaussian method of Atas & Bogomolny [2], which exploits the fact that the JW transformation maps the energy to a sum of independent fermionic occupation numbers. Grouping states by the total number of excited fermionic modes n , the energy distribution within each sector is approximately Gaussian by the central limit theorem, with mean and variance (Eqs. (32)–(33) of Ref. [2])

$$E_n = \left(\frac{2n}{N} - 1 \right) \langle \epsilon \rangle, \quad \sigma_n^2 = 4 \frac{n(N-n)}{N} (\langle \epsilon^2 \rangle - \langle \epsilon \rangle^2), \quad (2.5)$$

where $\langle \epsilon^k \rangle = \frac{1}{2\pi} \int_0^{2\pi} e(\phi)^k d\phi$ are moments of the single-mode dispersion

$$e(\phi) = \sqrt{1 + \lambda^2 - 2\lambda \cos \phi}, \quad \lambda = J_{zz}/h_x.$$

The total density of states is then

$$g(E) = \frac{1}{2^N} \sum_{n=0}^N \binom{N}{n} \frac{1}{\sqrt{2\pi} \sigma_n} \exp\left(-\frac{(E - E_n)^2}{2\sigma_n^2}\right). \quad (2.6)$$

Unlike the saddle-point approximation, the multi-Gaussian formula resolves the discrete sector structure of the spectrum at finite N . In particular, when the single-mode energies ϵ_k are bimodally distributed, which occurs when $J_{zz} \sim h_x$, as in our case, the multi-Gaussian approximation captures the resulting two-peak structure of $g(E)$, while the saddle-point approximation produces a single smooth curve. For this reason, the multi-Gaussian approximation will be used throughout this work when evaluating $p(E)$ analytically.

2.2 Purifications

A very common technique to address the many-body problem is purifications [3]: given a state ρ_S of a quantum system, it is possible to introduce an ancillary system $\mathcal{H}_A$ and define a pure state $|\Psi\rangle_{SA}$ in the joint system SA such that $\rho_S = \text{tr}_A(|\Psi\rangle\langle\Psi|)$. Purification offers a way around the many-body problem precisely because, for moderately large systems, the spectrum of ρ_S is inaccessible: working with the equivalent pure state $|\Psi\rangle_{SA}$ avoids ever having to diagonalize ρ_S directly, at the cost of finding ourselves in a larger-dimensional Hilbert space. The use of purifications and ancillas has countless applications, from quantum information theory to the problem at hand: the thermofield.

The thermofield [4] is a general approach that purifies a thermal state $\rho_\beta = e^{-\beta\hat{H}} / \text{tr}(e^{-\beta\hat{H}})$ in a doubled Hilbert space $\mathcal{H} \otimes \tilde{\mathcal{H}}$. The resulting pure state, known as the thermofield double state (TFD), admits an explicit representation in the eigenbasis of $\hat{H}$: if $\hat{H}|n\rangle = E_n|n\rangle$, then

$$|\Omega(\beta)\rangle = \frac{1}{\sqrt{Z(\beta)}} \sum_n e^{-\beta E_n/2} |n\rangle \otimes |\tilde{n}\rangle, \quad Z(\beta) = \sum_n e^{-\beta E_n}, \quad (2.7)$$

where $|\tilde{n}\rangle$ denotes the ancillary copy of the eigenstate $|n\rangle$. In this way, the expectation value of any observable in the original thermal state is recovered as an expectation value in the pure state:

$$\langle\hat{O}\rangle_\beta = \langle\Omega(\beta)|\hat{O} \otimes \tilde{\mathbb{I}}|\Omega(\beta)\rangle. \quad (2.8)$$

In compact notation, this state can be written as

$$|\Omega(\beta)\rangle \propto e^{-\frac{\beta}{2}\hat{H} \otimes \tilde{\mathbb{I}}} |I\rangle, \quad (2.9)$$

where $|I\rangle = \sum_n |n\rangle \otimes |\tilde{n}\rangle$ is the (unnormalized) state that maximally entangles each physical eigenstate with its ancillary copy.

In this work, the purification we construct does not aim to reproduce the thermofield double exactly. Instead, the ancilla is introduced so that its entanglement with the physical system, quantified by the Rényi-2 entropy S_2 , reproduces the entropic contribution to the free energy that would otherwise require diagonalizing ρ_S . This is developed in detail in the following section.

2.3 Rényi entropies, Maximal Rényi Ensemble

As an alternative to the von Neumann entropy, the Rényi entropies are defined as

$$S_\alpha = \frac{1}{1-\alpha} \ln \text{tr}(\rho^\alpha). \quad (2.10)$$

For $\alpha = 2$, this reduces to the Rényi-2 entropy:

$$S_2 = -\ln \text{tr}(\rho^2). \quad (2.11)$$

Compared to the von Neumann entropy, S_2 is computationally more accessible as will be explained later. Minimizing the functional $F_\alpha = \langle H \rangle - TS_\alpha$ over mixed states defines the so-called *maximal Rényi ensemble* (MRE) [5].

In this work, we exploit the fact that the MRE can be reformulated as a pure-state problem via the purification introduced in Section 2.2. Specifically, we introduce an ancillary system A of size N_A , and seek a pure state $|\Psi(\beta)\rangle$ on the combined system SA , such that the reduced state on the physical system S , of size N_S , approximates the MRE. The Hamiltonian of the joint system is taken to act trivially on the ancilla,

$$\hat{H}_{SA} = \hat{H}_S \otimes \mathbb{I}_A, \quad (2.12)$$

where $\hat{H}_S$ is the system Hamiltonian given by equation (2.1). The variational free energy to be minimized then takes the form:

$$F_2 = \langle \hat{H}_{SA} \rangle - TS_2, \quad S_2 = -\ln \text{tr}(\rho_S^2), \quad (2.13)$$

where $\rho_S = \text{tr}_A(|\Psi\rangle\langle\Psi|)$ is the reduced density matrix of the system (or equivalently of the ancilla, by symmetry), and is precisely the Rényi-2 entanglement entropy between system and ancilla. By construction $\langle \hat{H}_{SA} \rangle = \langle \hat{H}_S \rangle$, so the ancilla only intervenes in the entropic term, not in the energetic one; the temperature T enters explicitly in the definition of the free energy. By minimizing F_2 over pure states $|\Psi\rangle$, we obtain a variational purification of the MRE.

As discussed in Section 2.4, recovering the MRE at arbitrary temperatures requires an ancilla as large as the system ($N_A = N_S$). However, at sufficiently low temperatures, smaller ancillas ($N_A < N_S$) can still reproduce the thermal properties while reducing the computational cost.

The MRE admits an analytical solution in the eigenbasis of $\hat{H}_S$. To find it, one maximizes

the Rényi- α entropy subject to the constraints $\text{tr } \rho = 1$ and $\text{tr}(H\rho) = \bar{E}$, introducing Lagrange multipliers β_α and γ_α . At the stationary point, γ_α can be eliminated [6], and the optimal state takes the form:

$$\rho_\alpha = \frac{1}{Z_\alpha} \Pi_{E_\perp} \left(1 - \beta_\alpha \frac{\alpha-1}{\alpha} (H - \bar{E}) \right)^{\frac{1}{\alpha-1}} \Pi_{E_\perp}, \quad (2.14)$$

where $\Pi_{E_\perp} = \Theta(E_\perp - H)$ is a projector onto eigenstates with energy below the cutoff $E_\perp := \frac{\alpha}{\beta(\alpha-1)} + \bar{E}$, and Z_α is a normalization factor. This solution has two remarkable properties. First, the MRE spectrum exhibits an energy cutoff: unlike the Gibbs ensemble, where all eigenstates have weight $e^{-\beta E_k} > 0$, the MRE assigns zero weight to eigenstates with $E_k > E_\perp$. Second, the weight of each active eigenstate follows a power-law decay towards the cutoff, $(E_\perp - E_k)^{\frac{1}{\alpha-1}}$, instead of decaying exponentially.

For the case $\alpha = 2$, which is the one used in this work, the expression simplifies and the weight of the k -th level p_k is simply proportional to $(E_\perp - E_k)$. At zero temperature, the cutoff falls below the first excited level and the MRE reduces to the ground state. As T increases, the cutoff rises progressively and new eigenstates enter the ensemble discontinuously when $E_\perp$ crosses each eigenvalue, which results in changes in slope of local observables such as $\langle H \rangle$ or S_2 .

Despite this structural difference with respect to the Gibbs ensemble, Giudice *et al.* show that, in the thermodynamic limit, both ensembles yield the same expectation values of local observables when compared at the same mean energy $\bar{E}$, rather than at the same temperature T [6]. This distinction is crucial, since the relationship between T and $\bar{E}$ differs between the two ensembles, so fixing the temperature instead of the energy can in principle yield different results, even for local observables. The argument for the equivalence is based on the fact that the density of states $D(E)$ of a local Hamiltonian grows with N such that both distributions become increasingly concentrated around the same mean energy $\bar{E}$, with relative fluctuations of order $\mathcal{O}(1/\sqrt{N})$. Figure 2.1 illustrates this convergence numerically for different system sizes. For finite N , the MRE produces a piecewise linear dependence of observables on $\langle H \rangle/N$, a direct consequence of the linear form of the weights $p_k \propto \max(0, E_\perp - E_k)$ and the discrete set of active eigenstates. As N increases, the pieces become more numerous and the curve approaches the smooth result of the Gibbs ensemble. Furthermore, for $\alpha = 2$ it is observed numerically that, as $N \rightarrow \infty$, the parameter β_R that minimizes F_2 converges to the inverse temperature of the Gibbs ensemble with the same mean energy, which justifies interpreting T directly as the physical temperature in this limit.

It is worth noting that similar algorithms have been proposed in the literature [6, 7], but

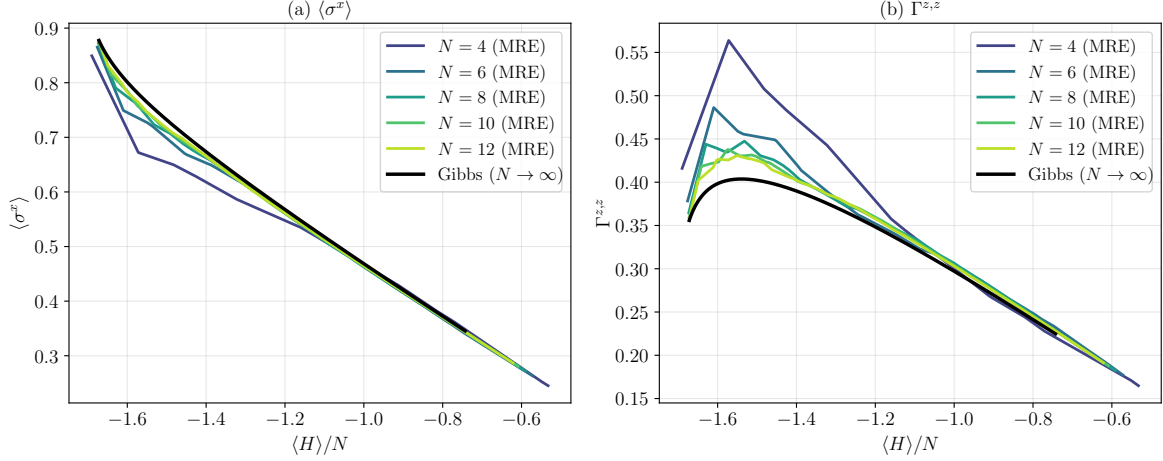


Figure 2.1: Convergence of local observables in the MRE to the thermodynamic limit of the canonical ensemble. The magnetization $\langle \sigma^x \rangle$ (a) and the next-neighbor correlation $\Gamma^{z,z}$ (b) obtained via ED for the MRE are compared with the analytical solution in the thermodynamic limit of the transverse field Ising model [1]. The Hamiltonian parameters are $J_{zz} = -1$ and $h_x = -1.5$.

these works study the Rényi ensemble using mixed states, rather than pure states arising from a purification as proposed in the present work.

2.4 Ancilla size

The size of the ancilla N_A plays a fundamental role in the ability of the method to represent thermal states at different temperatures. The maximum entanglement entropy that the ancilla system can store is bounded by

$$S_{\max} = N_{\min} \ln 2, \quad (2.15)$$

where $N_{\min} = \min(N_S, N_A)$ is the minimum between the number of qubits in the system and the ancilla. Therefore, for the method to be able to faithfully reproduce the thermal state at a given temperature, the second-order Rényi entanglement entropy S_2 must satisfy

$$S_2 \leq N_A \ln 2. \quad (2.16)$$

When $N_A = N_S$, the ancilla system has sufficient capacity to store all the required entanglement entropy at any temperature, thus recovering the MRE. However, if $N_A < N_S$, the ancilla imposes a limit on the maximum representable entropy, which restricts the range of accessible temperatures: the method can only reproduce the thermal state up to a certain temperature T^* above which S_2 exceeds the bound $N_A \ln 2$.

This behavior is illustrated in Figure 2.2, which shows the free energy, energy, and entanglement entropy as a function of temperature for different values of N_A (these are variational results obtained by training an NQS at each temperature, as explained in detail in the following section). It is observed that as N_A increases, the method is able to correctly reproduce the observables up to progressively higher temperatures, converging to the MRE result when $N_A = N_S$. Similarly, Figure 2.3 shows the expectation values $\langle \sigma^x \rangle$, $\Gamma_{x,x}$ and $\Gamma_{z,z}$, confirming that the accuracy of the results improves systematically with ancilla size.

However, using $N_A < N_S$ is particularly useful in the low-temperature regime, where the entanglement entropy is small and the condition $S_2 \leq N_A \ln 2$ is satisfied without needing a full-size ancilla. This significantly reduces the computational cost of the method, by decreasing the dimension of the Hilbert space over which the variational optimization is performed. In practice, therefore, an efficient strategy is to use the smallest ancilla that guarantees $S_2 \leq N_A \ln 2$ in the temperature range of interest, reserving $N_A = N_S$ only when access to high temperatures is required.

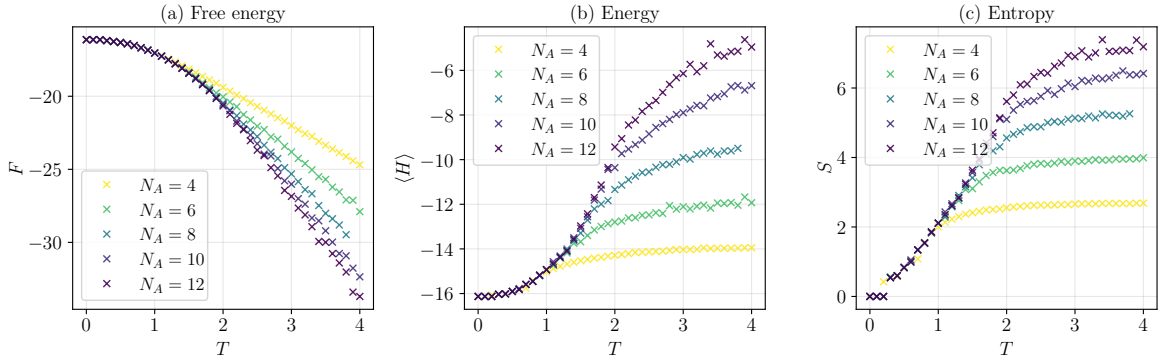


Figure 2.2: Free energy (a), energy (b), and entanglement entropy (c) as a function of temperature for different ancilla sizes N_A , with $N_S = 12$, $J_{ZZ} = -1$, $h_x = -1.1$. As N_A increases, the method correctly reproduces the thermal state up to progressively higher temperatures, recovering the MRE result when $N_A = N_S$.

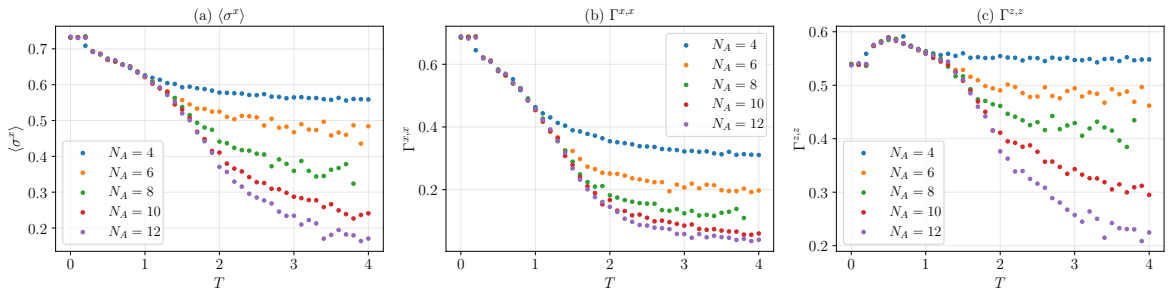


Figure 2.3: Expectation value of the x-axis magnetization $\langle \sigma^x \rangle$ (a), next-neighbor x-axis correlation $\Gamma_{x,x}$ (b) and next-neighbor z-axis correlation $\Gamma_{z,z}$ (c) as a function of temperature for different ancilla sizes N_A , with $N_S = 12$, $J_{ZZ} = -1$, $h_x = -1.1$.

2.5 Neural Quantum States

Representing a quantum state of N spin-1/2 particles requires, in principle, specifying 2^N complex amplitudes $\psi(\sigma)$. For $N = 20$, this is more than a million coefficients; for $N = 100$ it is an astronomical number ($2^{100} \approx 10^{30}$). Once the system is purified by appending an ancilla, this difficulty is compounded, since the relevant Hilbert space is that of the combined system SA .

A solution to this problem are Neural Quantum States (NQS) [8], in which, instead of parameterizing all amplitudes, they are parameterized by a neural network. In this way, the number of parameters scales polynomially with N , not exponentially.

An NQS expresses the wave function as

$$\psi_\theta(\sigma) = f_\theta(\sigma), \quad (2.17)$$

where f_θ is a neural network with parameters θ , which takes as input a configuration σ (encoded, for example, as $\sigma_i \in \{-1, +1\}$) and returns a complex number. In practice, it is common to directly parameterize the logarithm of the wave function.

One of the most important applications of NQS is the determination of the ground state (GS) of a Hamiltonian $\hat{H}$. To this end, a cost function that measures the expected energy is defined:

$$\mathcal{L}(\theta) = \langle \hat{H} \rangle_\theta = \sum_{\sigma} |\psi_\theta(\sigma)|^2 E_{\text{loc}}(\sigma), \quad (2.18)$$

where $E_{\text{loc}}(\sigma) = \frac{\langle \sigma | \hat{H} | \psi_\theta \rangle}{\psi_\theta(\sigma)}$ is the local energy. By minimizing $\mathcal{L}$, the network learns to approximate the ground state. This approach has proven to be extremely successful for a wide variety of systems, from spin chains to fermionic models. We can use this idea, and instead of minimizing H , minimize F_2 .

In this work we use the NetKet implementation [9], a library specialized in quantum many-body simulations based on JAX.

2.5.1 Autoregressive neural networks

In particular, we use autoregressive neural networks (ARNN), which parameterize the wave function as

$$\psi_\theta(\sigma) = \prod_{i=1}^{N_S+N_A} \sqrt{p_\theta(\sigma_i|\sigma_{<i})} e^{i\phi_\theta(\sigma_i|\sigma_{<i})}, \quad (2.19)$$

where $\sigma = (\sigma_1, \dots, \sigma_{2N})$ is a complete configuration (system + ancilla), $\sigma_{<i} = (\sigma_1, \dots, \sigma_{i-1})$ are the previous spins in a fixed order, and p_θ and ϕ_θ are the probability and phase learned by the network. The fundamental advantage of autoregressive networks is that they allow direct sampling: a configuration σ is generated sequentially, sampling $\sigma_1 \sim p_\theta(\cdot)$, then $\sigma_2 \sim p_\theta(\cdot|\sigma_1)$, and so on. This eliminates the autocorrelations typical of MCMC methods.

In our simulations we use the ARNNDense architecture from NetKet with the hyperparameters listed in Table 2.1.

Hyperparameter	Value
Number of hidden layers	1
Neurons per layer	16
Activation function	GELU (Gaussian Error Linear Unit)
Sampler	ARDirectSampler (direct autoregressive sampling)

Table 2.1: Hyperparameters of the ARNNDense architecture used in this work.

The network takes as input a configuration σ of the $2N$ spins (encoded as $\sigma_i \in \{-1, +1\}$) and outputs the logarithm of the wave function $\log \psi_\theta(\sigma)$.

2.5.2 Training and optimization

For each temperature T , we train the network by minimizing the Rényi-2 free energy:

$$F_2(\theta) = \langle \hat{H}_{SA} \rangle_\theta - T S_2(\theta), \quad (2.20)$$

where $\langle \hat{H}_{SA} \rangle_\theta$ is the expectation value of the total Hamiltonian over the purified state $|\Psi(\theta)\rangle$ and $S_2(\theta)$ is the Rényi-2 entropy of subsystem A computed from that state.

The gradient of F_2 with respect to the parameters θ decomposes as

$$\nabla_\theta F_2 = \nabla_\theta \langle \hat{H}_{SA} \rangle_\theta - T \nabla_\theta S_2(\theta), \quad (2.21)$$

where the first term is estimated using the standard variational Monte Carlo (VMC) estimator and the second term is obtained using the sampling estimator described in Appendix A. Both gradients are combined in a single backward pass implemented via the observable operator `FreeRenyiEnergyObservable`, which wraps the loss function F_2 as a NetKet observable and allows obtaining $\nabla_\theta F_2$ with a single call to `vstate.expect_and_grad`. The implementation

of the training loop is given in Appendix B.

Number of samples. We used $N_{\text{samples}} = 2^{18}$ samples for small systems ($N < 20$), while for large systems we used $N_{\text{samples}} = 2^{20}$. The large number of samples combats the noise problem of S_2 at high temperatures described in 3.2.1.

Optimizer. After comparing different strategies, we opted for `sign_sgd` from Optax with a learning rate that decays linearly from 0.05 to 0.001 over the training steps. Using ADAM produced similar results, while combining Stochastic Reconfiguration (SR) with SGD showed slower convergence and higher variance for this problem.

Stopping criterion and best state selection. Training is run for a fixed number of $N_{\text{steps}} = 350$ steps. Throughout training, F_2 is recorded at each step and the parameters θ^* corresponding to the achieved minimum are kept. At the end, these parameters are restored and $\langle \hat{H}_{SA} \rangle$ and S_2 are re-evaluated with independent high-statistics estimates to obtain the reported values.

Gradient estimator reliability. To validate the consistency of the stochastic estimator of $\nabla_{\theta} S_2$, the gradient computation is repeated $N_{\text{rep}} = 10$ times on the final state and the cosine similarity between all pairs of estimates is calculated. A value close to 1 indicates that the estimator has low variance and that the descent directions are consistent across independent replicas.

2.6 Rényi-2 entanglement entropy

The entanglement entropy is a fundamental measure that quantifies the quantum correlations between a subsystem and its complement. In the context of this work, we use it to characterize the purified state that describes the thermal system. As introduced in Section 2.3, we work with the Rényi-2 entropy defined as:

$$S_2 = -\ln \text{tr} \left(\text{tr}_A(\rho)^2 \right), \quad (2.22)$$

where ρ is the density matrix of the purified state (system + ancilla), tr_A denotes the partial trace over the ancilla, and the result is precisely the entanglement entropy between the system and the ancilla. Below we present the methods used to estimate this quantity.

2.6.1 Swap trick

Computing the Rényi-2 entropy requires estimating the purity $\text{tr}(\rho_A^2)$ of the reduced density matrix ρ_A of subsystem A . The swap trick [10] allows expressing this quantity as an expectation value accessible via Monte Carlo.

Consider two independent copies of the system, with configurations $\sigma = (\sigma_S, \sigma_A)$ and $\tau = (\tau_S, \tau_A)$, where the subscripts S and A denote system and ancilla respectively. We define the swap operator $SWAP_A$ that exchanges the ancilla configurations between the two copies:

$$SWAP_A |\sigma_S, \sigma_A\rangle \otimes |\tau_S, \tau_A\rangle = |\sigma_S, \tau_A\rangle \otimes |\tau_S, \sigma_A\rangle. \quad (2.23)$$

As shown in Appendix A.1, the expectation value of this operator in the product state $|\Psi\rangle \otimes |\Psi\rangle$ is exactly the purity:

$$\langle SWAP_A \rangle = \text{tr}(\rho_A^2). \quad (2.24)$$

In practice, we define the swap ratio

$$R = \frac{|\Psi(\sigma_S, \tau_A)|^2 |\Psi(\tau_S, \sigma_A)|^2}{|\Psi(\sigma_S, \sigma_A)|^2 |\Psi(\tau_S, \tau_A)|^2}, \quad (2.25)$$

where the numerator corresponds to the configurations after the swap and the denominator to the original ones. It can be shown that $\langle R \rangle = \text{tr}(\rho_A^2)$. Therefore,

$$S_2 = -\ln \langle R \rangle. \quad (2.26)$$

This formulation allows estimating S_2 by averaging the observable R over independent samples from the distribution $|\Psi|^2 \otimes |\Psi|^2$.

To minimize the free energy $F = \langle H \rangle - TS_2$ via gradient descent, we need to estimate $\nabla_\theta F$ with respect to the neural network parameters θ . The gradient of S_2 is obtained by differentiating the swap trick estimator, and its exact calculation is detailed in Appendix A.2. The numerical implementation is described in Appendix A.3.

2.6.2 λ -integral

Although the swap trick provides a simple expression for S_2 , its Monte Carlo estimation suffers from a signal-to-noise problem when S_2 is large. In that regime, $\langle R \rangle$ is very small, and most samples have $R \approx 0$, with only a fraction $\sim e^{-S_2}$ contributing significantly.

To mitigate this problem, Drut and Porter [11] proposed a method based on an integral over an auxiliary parameter λ . The idea is to define

$$\Gamma(\lambda) = \int \mathcal{D}\{\sigma\} P[\{\sigma\}] Q^\lambda[\{\sigma\}], \quad (2.27)$$

where Q is the entanglement determinant (analogous to R in our context). It holds that

$$\ln \Gamma(0) = 0, \quad \ln \Gamma(1) = (1 - n)S_n. \quad (2.28)$$

Differentiating with respect to λ ,

$$\frac{\partial \ln \Gamma}{\partial \lambda} = \langle \ln Q \rangle_\lambda, \quad (2.29)$$

where $\langle \cdot \rangle_\lambda$ denotes the average with respect to the measure $\tilde{P}_\lambda \propto P Q^\lambda$. Integrating from $\lambda = 0$ to $\lambda = 1$,

$$S_n = \frac{1}{1 - n} \int_0^1 d\lambda \langle \ln Q \rangle_\lambda. \quad (2.30)$$

The advantage of this formulation is that the integral averages out statistical fluctuations along the path in λ . In particular, for $\lambda = 0$ the measure is the original one, while for $\lambda = 1$ the measure is biased towards configurations with large Q . The integration yields a more stable estimate.

The authors observe that $\langle \ln Q \rangle_\lambda$ is a smooth and nearly linear function of λ (see Fig. 2 of [11]), which allows performing the numerical integration with a uniform grid of few points. Furthermore, they note that the distribution of Q is approximately log-normal (Fig. 1 of [11]), which explains the effectiveness of the method.

3. Results

3.1 Validation

As discussed in Section 2.3, the MRE admits an analytical solution. For the case $\alpha = 2$, this solution takes the form:

$$\rho_2 = \frac{1}{Z_2} \Pi_{E_\perp} \left(1 - \frac{\beta_2}{2} (H - \bar{E}) \right) \Pi_{E_\perp}. \quad (3.1)$$

This expression allows computing expectation values of observables via exact diagonalization in small systems, serving as a benchmark for the variational results obtained with the method of Chapter 2. Additionally, we can determine the temperatures T_n at which each new eigenstate of $\hat{H}_S$ becomes active. Following the procedure in Appendix C, we obtain:

$$T_n = \frac{1}{2} \left(E_n - \frac{\sum_{k=0}^{n-1} E_k (E_n - E_k)}{\sum_{k=0}^{n-1} (E_n - E_k)} \right). \quad (3.2)$$

Figure 3.1 shows the results obtained via exact diagonalization (ED) and variationally, together with the canonical ensemble solution (also obtained via ED) for the case $N = 2$, and Figure 3.2 for the case $N = 4$. In both cases we take $J_{zz} = -1$, $h_x = -1.5$.

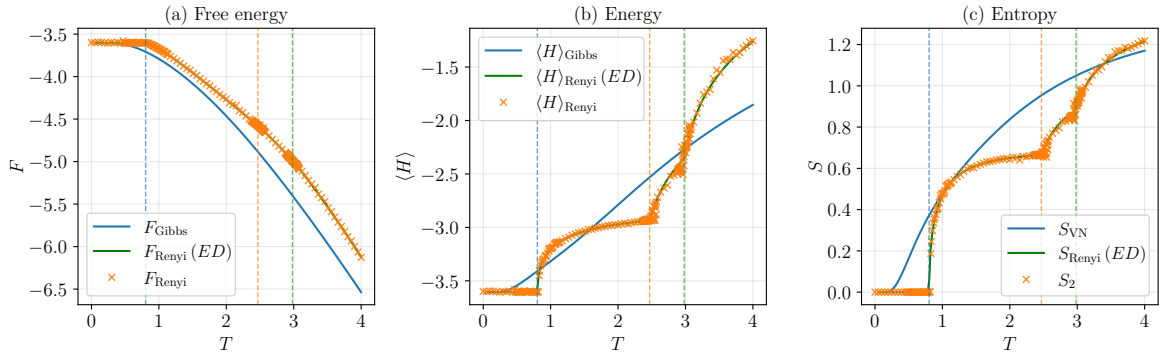


Figure 3.1: Free energy (a), energy (b), and entropy (c) as a function of temperature for $N = 2$, $J_{zz} = -1$, $h_x = -1.5$. The MRE solution obtained via ED is shown (green line), variationally (orange crosses), and the canonical ensemble solution (blue line) for comparison. The vertical dotted lines indicate the temperatures T_n at which a new eigenstate becomes active, given by equation 3.2.

As observed in Figures 3.1 and 3.2, the variational results coincide with the analytical solution for the MRE, and the minimum of the free energy $F_2 = \langle H \rangle - TS_2$ is correctly found. We can also verify that training the network yields the same solution as ED for local

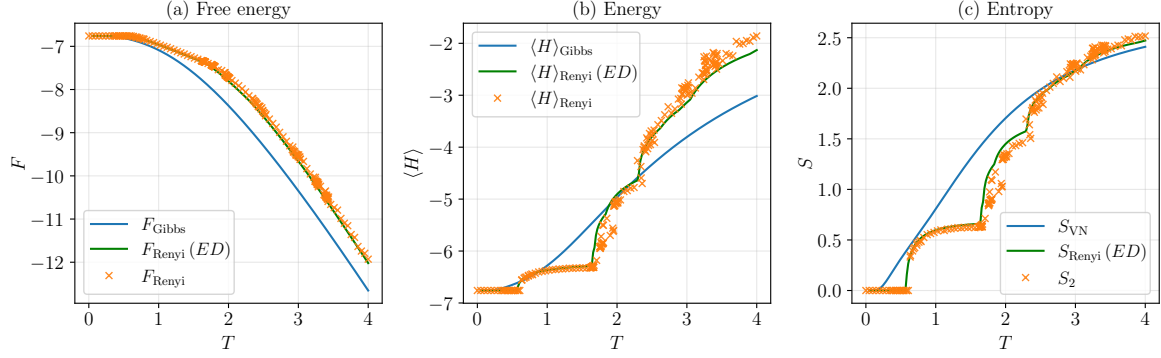


Figure 3.2: Free energy (a), energy (b), and entropy (c) as a function of temperature for $N = 4$, $J_{zz} = -1$, $h_x = -1.5$. The MRE solution obtained via ED is shown (green line), variationally (orange crosses), and the canonical ensemble solution (blue line) for comparison.

observables such as magnetization or nearest-neighbor correlations. Figures 3.3 and 3.4 show the expectation values of the x-magnetization and x and z next-neighbor correlations as a function of temperature for the previously trained networks.

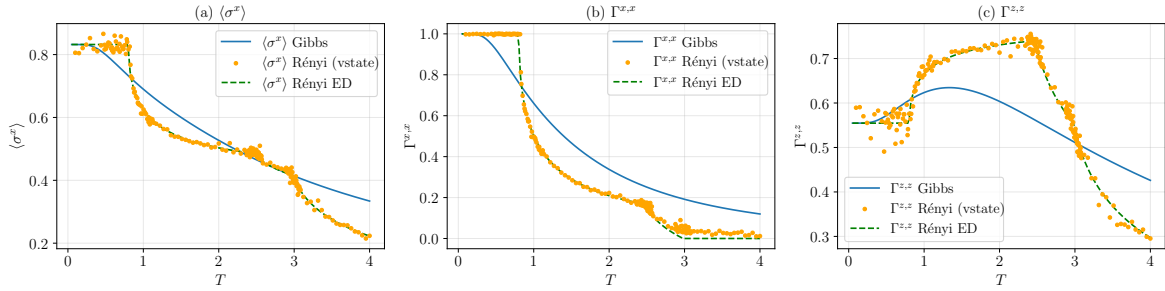


Figure 3.3: Values of $\langle \sigma^x \rangle$ (a), $\Gamma^{x,x}$ (b) and $\Gamma^{z,z}$ (c) as a function of temperature for $N = 2$, $J_{zz} = -1$, $h_x = -1.5$. Variational results (orange points) are compared with the exact MRE (green line) and the Gibbs ensemble (blue line).

3.2 Scaling with system size

Once the correct convergence of the neural network training for small systems and the correct reproduction of the MRE ground state have been verified, the next natural step is to increase the system size. In the thermodynamic limit, the local properties of the MRE are expected to converge to those of the Gibbs ensemble, as discussed in Section 2.3. Studying larger systems allows us to explore this equivalence and analyze the performance of the variational method on scales where exact diagonalization becomes unfeasible.

However, increasing N also increases the maximum entanglement entropy S_2 that the system can develop at a given temperature, since S_2 is an extensive quantity bounded by $N \ln 2$. As discussed below, the statistical error of the swap-trick and λ -integral estimators

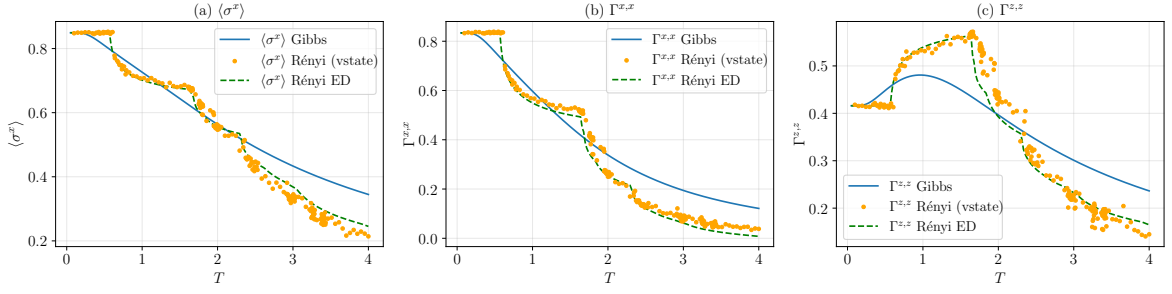


Figure 3.4: Expectation values of $\langle \sigma^x \rangle$ (a), $\Gamma^{x,x}$ (b) and $\Gamma^{z,z}$ (c) as a function of temperature for $N = 4$, $J_{ZZ} = -1$, $h_x = -1.5$. Variational results (orange points) are compared with the exact MRE (green line) and the Gibbs ensemble (blue line).

grows exponentially with S_2 , so larger systems are intrinsically more affected by this noise problem at a fixed temperature. This trade-off between system size and estimator reliability is a recurring theme throughout this section.

3.2.1 Noise in S_2

Despite the advantages of the λ -integral method, the noise problem does not completely disappear when the entropy is high. The fundamental reason is that the estimator still depends on the entanglement determinant Q (or R in our case), whose distribution has long tails. In the high-entropy limit, only a fraction $\sim e^{-S_2}$ of samples contributes significantly to the average $\langle R \rangle$, which causes the estimates of S_2 and, more critically, of its gradient, to be very noisy. Concretely, the statistical error in S_2 grows exponentially with entropy, following the theoretical trend

$$\sigma_{S_2} \sim \frac{e^{S_2/2}}{\sqrt{N_{\text{samples}}}}, \quad (3.3)$$

so that the number of samples required to maintain a fixed error scales as e^{S_2} .

We have quantified this effect for our spin model. To this end, networks were trained at different temperatures, which correspond to different entanglement entropies: since S_2 grows with T , a sweep in temperature effectively produces a sweep in S_2 . We restrict this analysis to a small system size ($N = 6$), which allows us to obtain both S_2 and the exact gradient of S_2 with respect to the network parameters via exact diagonalization, providing a ground truth against which the sampling-based estimators of S_2 can be compared. Figure 3.5 shows the absolute and relative error in S_2 as a function of the exact entropy. Both methods (standard swap trick and λ -integral) produce very similar results, with an error that grows exponentially with S_2 and whose trend is well described by the theoretical line of equation (3.3).

Figure 3.6 shows the gradient quality, measured by the cosine between the estimated gradient and the exact one (obtained by exact diagonalization for $N = 6$). For low entropies

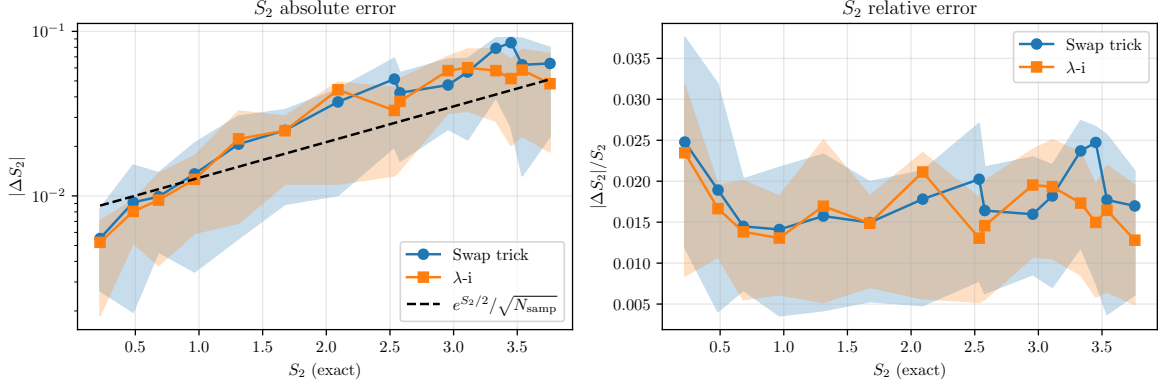


Figure 3.5: Absolute error (left) and relative error (right) in S_2 as a function of the exact entropy. The bands represent the 25–75 percentiles over 50 repetitions. The dashed line shows the theoretical trend $e^{S_2/2}/\sqrt{N_{\text{samples}}}$.

($S_2 \lesssim 2$), both methods achieve cosines close to 1. As S_2 increases, the cosine drops dramatically, reaching values ~ 0.6 for $S_2 \approx 3.5$, evidencing that the estimated descent directions are unreliable in this regime.

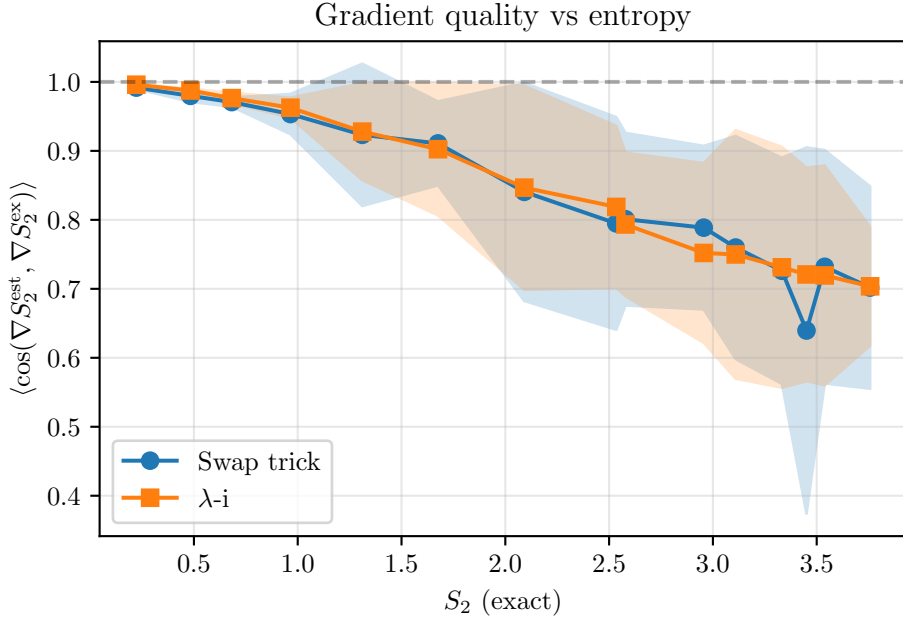


Figure 3.6: Cosine between the estimated gradient and the exact gradient as a function of S_2 . The bands represent the 25–75 percentiles over 50 repetitions.

We have also analyzed the dependence on the number of samples. Figures 3.7a and 3.7b show the absolute error in S_2 for low entropy ($S_2 \approx 1.0$) and high entropy ($S_2 \approx 3.5$), respectively. For low entropy, the error scales as $1/\sqrt{N_{\text{samples}}}$ and converges quickly to zero. For high entropy, in contrast, orders of magnitude more samples are needed to achieve the same precision, consistent with the exponential scaling of equation (3.3).

Regarding gradient quality, Figures 3.8a and 3.8b show the cosine as a function of the

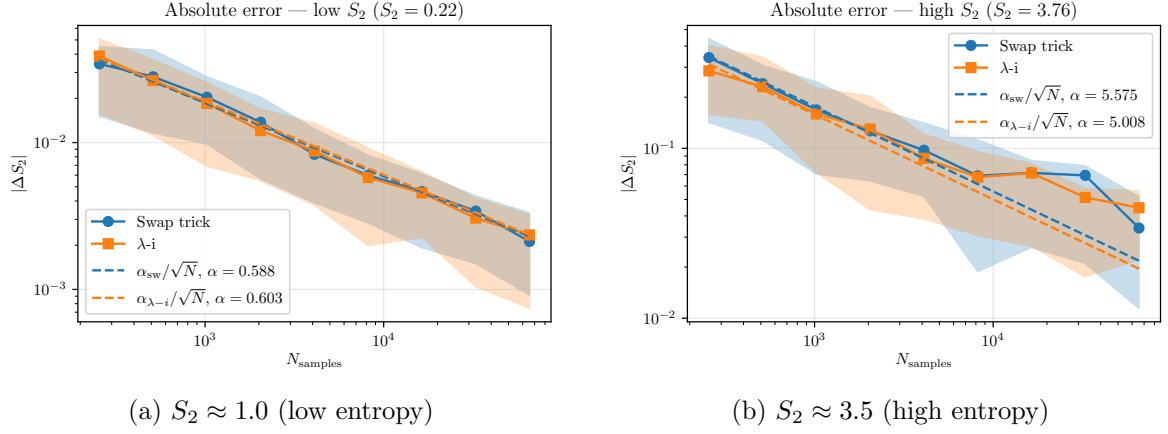


Figure 3.7: Absolute error in S_2 as a function of the number of samples for (a) low entropy and (b) high entropy. The dashed lines show the fit $\alpha/\sqrt{N_{\text{samples}}}$. The bands represent the 25–75 percentiles over 50 repetitions.

number of samples. For low entropy, a few thousand samples already yield a cosine close to 1, and the λ -integral offers a marginal improvement over the standard swap trick. For high entropy, even with $N_{\text{samples}} = 65536$ the cosine does not exceed ~ 0.85 , highlighting that neither method solves the underlying problem in this regime.

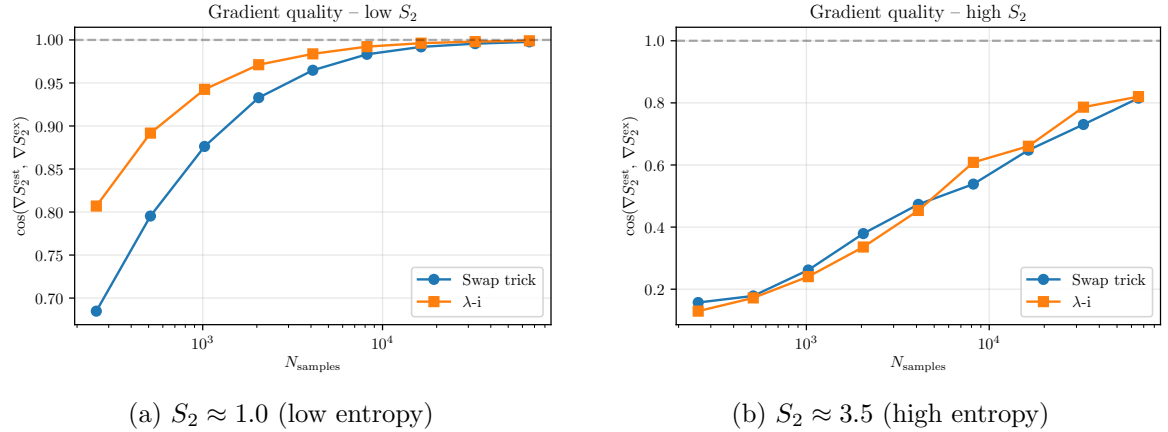


Figure 3.8: Gradient cosine as a function of the number of samples for (a) low entropy and (b) high entropy.

In summary, the main limitation is statistical in nature: the rarity of configurations relevant for $\langle R \rangle$ makes the sample cost grow exponentially with S_2 . The λ -integral partially mitigates the problem by redistributing statistical weight along the integration path, but the point $\lambda = 1$ remains the most critical and cannot be avoided. For systems with high entanglement entropy (such as those appearing at high temperatures or with large subsystems) this constitutes a fundamental limitation of the method that should be taken into account when interpreting training results. To characterize up to what entropy, and therefore temperature, the results of a training are reliable, one takes 10 batches of samples and computes the mean

cosine of the gradient of S_2 between them, ensuring that it does not fall below a critical value (typically 0.7).

3.2.2 Convergence of local observables

Figure 3.9 shows the expectation values of two local observables, $\langle \sigma^x \rangle$ and $\Gamma^{z,z}$, as a function of the mean energy per site $\langle H \rangle / N$, for system sizes $N = 15, 20, 30, 50$, alongside the thermodynamic limit (T.L.) prediction of the Gibbs ensemble. These results correspond to converged variational trainings, and are subject to the limitations imposed by the noise in S_2 discussed in Section 3.2.1: for $N = 50$, in particular, it was not possible to reach higher energies per site, since the corresponding entropy exceeds the regime where the gradient estimator remains reliable. Nonetheless, these are numerical results for system sizes inaccessible to exact diagonalization, and as predicted by Giudice et al. [6], local observables in the MRE converge to their Gibbs ensemble counterparts in the thermodynamic limit: our results are consistent with this prediction, since as N increases, the variational MRE results approach the thermodynamic limit Gibbs curve for both observables.

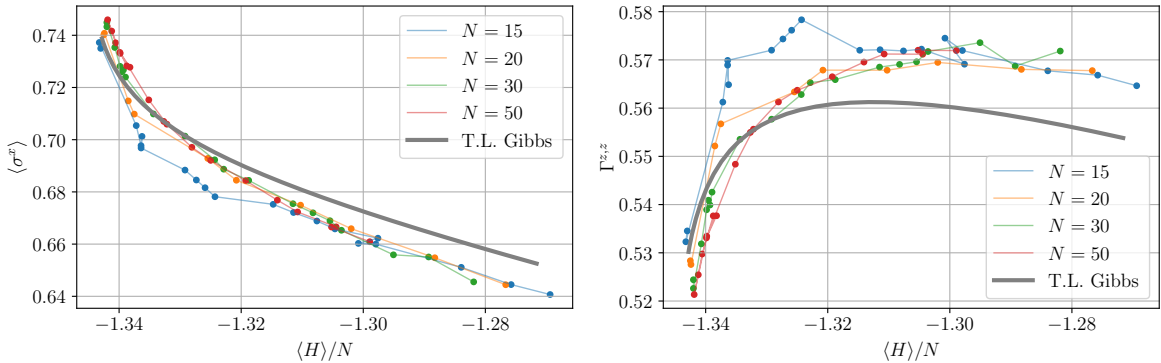


Figure 3.9: Expectation values of $\langle \sigma^x \rangle$ (left) and $\Gamma^{z,z}$ (right) as a function of mean energy per site $\langle H \rangle / N$ for $N = 15, 20, 30, 50$, $J_{ZZ} = -1$, $h_x = -1.1$. Variational results are compared with the Gibbs ensemble thermodynamic limit (gray line).

The theoretical basis for this equivalence rests on a statistical argument. For local Hamiltonians, the density of states $D(E)$ becomes Gaussian in the thermodynamic limit as a consequence of the central limit theorem. As a result, both the Gibbs and the Rényi energy distributions become increasingly narrow and peaked around the same mean energy $\bar{E}$, with fluctuations that scale as $\mathcal{O}(1/\sqrt{N})$. Any local observable that varies smoothly with the energy must therefore take the same value in both ensembles as $N \rightarrow \infty$. It is important to note, however, that this equivalence is established by comparing the two ensembles at the same mean energy $\bar{E}$, not at the same temperature T . This distinction is crucial: the relationship between T and $\bar{E}$ differs between the Gibbs and Rényi ensembles, so fixing the

temperature instead of the energy can yield very different results, even for local observables.

This subtlety is precisely what reconciles the prediction of Giudice et al. [6] with the seemingly contradictory findings of Jreissaty et al. [12]. In the latter work, the two ensembles are compared at fixed temperature T , not at fixed energy, and the critical temperature T_c of the 2D classical Ising model is found to depend strongly on the Rényi index α . The magnetization $|m|$ as a function of T differs clearly between the Rényi and Gibbs ensembles, even at large system sizes and even for this simple local observable. This is not in contradiction with Giudice et al., but rather a consequence of the fact that the same temperature T corresponds to different mean energies in the two ensembles, so one is not comparing equivalent physical situations. Jreissaty et al. themselves acknowledge that their results near the critical point do represent a genuine departure from the claim of local indistinguishability, and identify the origin of this discrepancy: near a phase transition, the divergence of the correlation length ξ means that no observable is truly local in the thermodynamic sense, and the Gaussian approximation for the density of states breaks down precisely in this regime. The argument of Giudice et al. therefore holds away from criticality, but the Rényi ensemble genuinely deforms the critical physics, shifting T_c and, at the mean-field level, even changing the order of the phase transition above a threshold value of α .

In the one-dimensional quantum Ising model studied in this work, magnetic long-range order exists only at $T = 0$, and no finite-temperature phase transition occurs [1, Fig. 4.3], so the conditions for ensemble equivalence are satisfied at all temperatures considered. The convergence observed in Figure 3.9 is therefore expected to persist as $N \rightarrow \infty$, and the results confirm that the MRE provides a reliable approximation to the thermal state for local observables in this setting.

3.2.3 Energy probability distribution

A deeper understanding of the convergence mechanism can be obtained by examining the energy probability distribution $p(E)$, which encodes how the two ensembles weight the different energy sectors of the spectrum. In the Gibbs ensemble, the distribution is given by

$$p_{\text{Gibbs}}(E) \propto g(E) e^{-\beta E}, \quad (3.4)$$

where $g(E)$ is the density of states. In the MRE with $\alpha = 2$, the corresponding distribution is (Eq. (2) of Ref. [12])

$$p_{\text{Rényi}}(E) \propto g(E) \left[1 - \frac{\beta}{2} N \left(\frac{E}{N} - \bar{E} \right) \right], \quad (3.5)$$

valid only for energy levels satisfying the accessibility condition (Eq. (4) of Ref. [12])

$$E \leq \bar{E} + \frac{2}{\beta}, \quad (3.6)$$

and zero otherwise. This condition ensures the positivity of $p_{\text{Rényi}}(E)$ and has no analogue in the Gibbs ensemble, where all energy levels remain accessible with exponentially suppressed but nonzero probability. The key structural difference between Eqs. (3.4) and (3.5) is that the Gibbs weight is an exponentially decaying function of energy, whereas the MRE weight is a linearly decreasing function. As a consequence, $p_{\text{Rényi}}(E)$ has a hard cutoff at the energy

$$E_{\text{cut}} = \bar{E} + \frac{2}{\beta}, \quad (3.7)$$

above which no states are accessible to the ensemble. This cutoff has no analogue in the Gibbs ensemble, where all energies remain accessible with exponentially suppressed but nonzero probability.

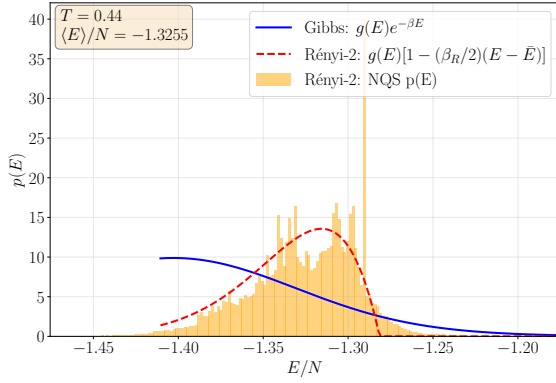
A further complication of the MRE is that $\bar{E}$ appears explicitly inside the distribution itself, leading to the self-consistency condition

$$\bar{E} = \int E p_{\text{Rényi}}(E | \bar{E}) dE, \quad (3.8)$$

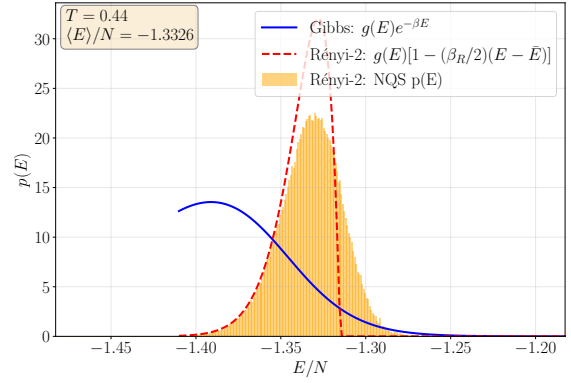
which must be solved iteratively. As shown analytically by Jreissaty et al. [12] for the one-dimensional Ising model (Section IV of that reference), this fixed-point equation can admit multiple solutions at a given temperature, each corresponding to a different self-consistent energy sector of the spectrum. In our numerical calculations, we identify this solution directly from the trained variational state by setting $\bar{E} = \langle H \rangle_{\text{NQS}}$, which by construction satisfies the self-consistency condition at the free energy minimum reached during training.

Figures 3.10, 3.11 show the energy distribution $p(E)$ for different system sizes and temperatures. Three curves are displayed in each panel: the Gibbs distribution (3.4), the analytical MRE distribution (3.5) evaluated at $\bar{E} = \langle H \rangle_{\text{NQS}}$, and the histogram of local energy estimators sampled from the variational state, which provides a direct empirical estimate of $p_{\text{Rényi}}(E)$.

For $N = 20$ (Figure 3.10a), the distributions of the Gibbs and MRE ensembles are visibly distinct: they differ both in width and in shape. The histogram of the NQS samples agrees well with the analytical MRE curve, confirming that the variational state has correctly learned the MRE distribution. Notably, the NQS histogram displays sharp peaks that are not present in the smooth analytical curve. These peaks reflect the discrete structure of the density of

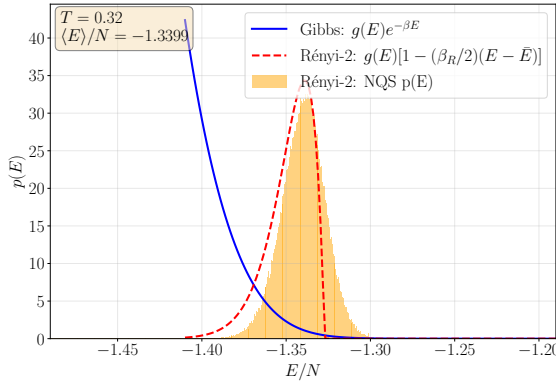


(a) $N = 20$

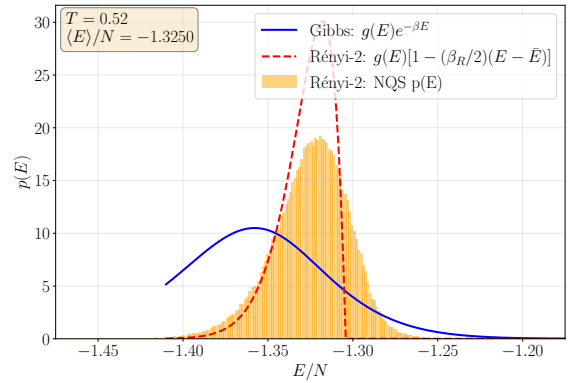


(b) $N = 50$

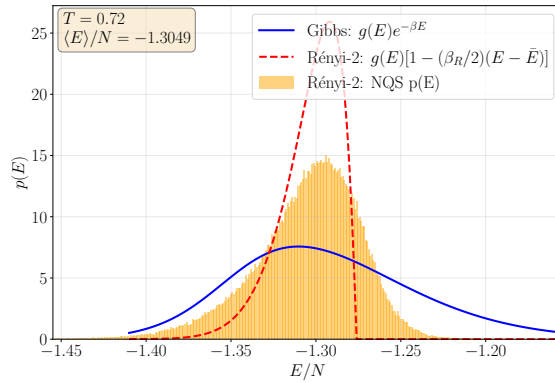
Figure 3.10: Probability density $p(E)$ as a function of energy per site E/N for $T = 0.44$, $J_{ZZ} = -1, h_x = -1.1$, with $N = 20$ (a) and $N = 50$ (b). Blue line corresponds to the Gibbs ensemble (analytical), red dashed line to the Rényi ensemble (analytical), and orange bars to the Rényi ensemble (NQS results).



(a) $T = 0.32$



(b) $T = 0.52$



(c) $T = 0.72$

Figure 3.11: Probability density $p(E)$ as a function of energy per site E/N for $T = 0.32$ (a), $T = 0.52$ (b) and $T = 0.72$ (c), $J_{ZZ} = -1, h_x = -1.1$, with $N = 50$. Blue line corresponds to the Gibbs ensemble (analytical), red dashed line to the Rényi ensemble (analytical), and orange bars to the Rényi ensemble (NQS results).

states $g(E)$ at finite N : even though the analytical approximation appears smooth, this is an artifact of the approximation used to compute $g(E)$, which becomes exact only in the thermodynamic limit. The true $g(E)$ at $N = 20$ has a discrete multi-Gaussian structure (as discussed below), and the NQS samples resolve this fine structure.

As N increases to 50 (Figure 3.10a), both distributions become significantly narrower, as expected from the $1/\sqrt{N}$ scaling of energy fluctuations in extensive systems. The Gibbs and MRE distributions approach each other and become increasingly centered at the same mean energy. This narrowing is the microscopic mechanism behind the ensemble equivalence for local observables: when $p(E)$ is sufficiently narrow, any smooth observable $O(E)$ satisfies

$$\langle O \rangle = \int O(E) p(E) dE \approx O(\bar{E}), \quad (3.9)$$

independently of the precise shape of $p(E)$, so that the Gibbs and MRE ensembles yield the same expectation values. In the microcanonical sense, both ensembles converge to the same energy shell as $N \rightarrow \infty$.

It is also observed that at lower temperatures the NQS distribution matches the analytical MRE curve more accurately, while at higher temperatures the agreement deteriorates and the NQS histogram becomes broader. This is likely due to the increasing difficulty of estimating the gradient of the second Rényi entropy S_2 at high temperatures (as discussed in Section 3.2.1), hindering convergence of the variational optimization.

4. Conclusions and outlook

In this work, we have developed a variational framework based on Neural Quantum States and purifications to approximate finite-temperature states through the minimization of the Rényi-2 free energy. The method reformulates the search for the maximal Rényi ensemble (MRE) as a pure-state optimization problem, where the thermal entropy is replaced by the Rényi-2 entanglement entropy between a physical system and an ancillary Hilbert space.

The approach was implemented using autoregressive neural networks within the NetKet framework and validated against exact diagonalization results for small systems. In all tested cases, the variational optimization successfully reproduced the analytical MRE solution, demonstrating that the free-energy minimum can be located accurately using stochastic gradient-based methods.

One of the main findings of this work is that the size of the ancillary system does not necessarily need to match the size of the physical system. While recovering the exact MRE over the entire temperature range requires an ancilla of equal size, we found that at low temperatures a significantly smaller ancilla can already reproduce the relevant thermal properties with good accuracy. This reduction in ancilla size leads to a substantial decrease in computational cost, making the method considerably more practical in the low-temperature regime.

The analysis of the gradient quality confirms that this issue constitutes the primary computational bottleneck of the method. Although the λ -integral formulation provides a modest improvement in some situations, it does not eliminate the fundamental exponential scaling associated with the sampling of rare but highly relevant configurations. Therefore, the accessible temperature range is determined not only by the expressive power of the neural-network ansatz, but also by the ability to estimate the entropic contribution with sufficient accuracy.

Several directions for future work remain open. In particular, it would be interesting to investigate alternative entropy estimators, improved sampling strategies, or importance-sampling schemes specifically designed to address the long-tail problem. Extending the method to more challenging quantum many-body systems, where conventional thermal-state approaches become difficult, also constitutes a promising avenue for future research.

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A. Rényi entropy calculations

A.1 Swap trick derivation

We now demonstrate that $\langle \text{SWAP}_A \rangle = \text{tr}(\rho_A^2)$. Starting from the expectation value of the operator SWAP_A in the product state $|\Psi\rangle \otimes |\Psi\rangle$:

$$\langle \text{SWAP}_A \rangle = \sum_{\substack{\sigma_S, \sigma_A \\ \tau_S, \tau_A}} |\Psi(\sigma_S, \sigma_A)|^2 |\Psi(\tau_S, \tau_A)|^2 \langle \sigma_S, \sigma_A; \tau_S, \tau_A | \text{SWAP}_A | \sigma_S, \sigma_A; \tau_S, \tau_A \rangle. \quad (\text{A.1})$$

The operator SWAP_A only exchanges the ancillas between the two copies,

$$\text{SWAP}_A | \sigma_S, \sigma_A; \tau_S, \tau_A \rangle = | \sigma_S, \tau_A; \tau_S, \sigma_A \rangle, \quad (\text{A.2})$$

so the matrix element reduces to

$$\langle \sigma_S, \sigma_A; \tau_S, \tau_A | \text{SWAP}_A | \sigma_S, \sigma_A; \tau_S, \tau_A \rangle = \delta_{\sigma_A, \tau_A}. \quad (\text{A.3})$$

Substituting and performing the sum over τ_A with the Kronecker delta:

$$\langle \text{SWAP}_A \rangle = \sum_{\sigma_S, \tau_S, \sigma_A} |\Psi(\sigma_S, \sigma_A)|^2 |\Psi(\tau_S, \sigma_A)|^2. \quad (\text{A.4})$$

We recognize in this expression the elements of the reduced density matrix $\rho_A(\sigma_A, \tau_A) = \sum_{\sigma_S} \Psi(\sigma_S, \sigma_A) \Psi^*(\sigma_S, \tau_A)$, whose squared trace is

$$\text{tr}(\rho_A^2) = \sum_{\sigma_A} [\rho_A^2]_{\sigma_A \sigma_A} = \sum_{\sigma_A, \tau_A} |\rho_A(\sigma_A, \tau_A)|^2 = \sum_{\sigma_S, \tau_S, \sigma_A} |\Psi(\sigma_S, \sigma_A)|^2 |\Psi(\tau_S, \sigma_A)|^2, \quad (\text{A.5})$$

where in the last step we have expanded $|\rho_A(\sigma_A, \tau_A)|^2 = \rho_A(\sigma_A, \tau_A) \rho_A^*(\sigma_A, \tau_A)$ and used the hermiticity of ρ_A to identify the resulting factors with $|\Psi(\sigma_S, \sigma_A)|^2$ and $|\Psi(\tau_S, \sigma_A)|^2$.

A.2 Gradient of the S_2 estimator

We start from the swap trick estimator $e^{-S_2} = \langle R \rangle_\theta$, where $\langle \cdot \rangle_\theta$ denotes the average over the sampling distribution $p_\theta(x) = |\Psi_\theta(x)|^2 / Z_\theta$. We want to compute $\nabla_\theta S_2$.

Exact gradient derivation

Differentiating $S_2 = -\ln\langle R \rangle_\theta$ with respect to θ :

$$\nabla_\theta S_2 = -\frac{\nabla_\theta \langle R \rangle_\theta}{\langle R \rangle_\theta}. \quad (\text{A.6})$$

The gradient of $\langle R \rangle_\theta = \sum_{x,y} p_\theta(x) p_\theta(y) R_\theta(x,y)$ has two contributions:

$$\nabla_\theta \langle R \rangle_\theta = \langle \nabla_\theta R_\theta \rangle_\theta + \sum_{x,y} R_\theta(x,y) \nabla_\theta [p_\theta(x) p_\theta(y)]. \quad (\text{A.7})$$

Since $\ln p_\theta(x) = 2 \operatorname{Re}(\ln \Psi_\theta(x)) - \ln Z_\theta$, we have

$$\nabla_\theta \ln p_\theta(x) = 2 O_\theta(x) - \nabla_\theta \ln Z_\theta, \quad O_\theta(x) \equiv \nabla_\theta \ln \Psi_\theta(x), \quad (\text{A.8})$$

where $\nabla_\theta \ln Z_\theta = 2\langle O_\theta \rangle_\theta$. Therefore,

$$\begin{aligned} \nabla_\theta [p_\theta(x) p_\theta(y)] &= p_\theta(x) p_\theta(y) [\nabla_\theta \ln p_\theta(x) + \nabla_\theta \ln p_\theta(y)] \\ &= p_\theta(x) p_\theta(y) \cdot 2(O_\theta(x) + O_\theta(y) - 2\langle O_\theta \rangle_\theta). \end{aligned} \quad (\text{A.9})$$

For the first term, using $R_\theta(x,y) = e^{\ln R_\theta(x,y)}$:

$$\nabla_\theta R_\theta = R_\theta \nabla_\theta \ln R_\theta = R_\theta (O_\theta(x') + O_\theta(y') - O_\theta(x) - O_\theta(y)) \cdot 2, \quad (\text{A.10})$$

where $x' = (x_S, y_A)$ and $y' = (y_S, x_A)$ are the swapped configurations. Letting $\Delta O_\theta \equiv O_\theta(x') + O_\theta(y') - O_\theta(x) - O_\theta(y)$, we have $\langle \nabla_\theta R_\theta \rangle_\theta = \langle R_\theta \cdot 2\Delta O_\theta \rangle_\theta$.

Combining both contributions:

$$\nabla_\theta \langle R \rangle_\theta = \langle R_\theta \cdot 2\Delta O_\theta \rangle_\theta + 2\langle R_\theta (O_\theta(x) + O_\theta(y) - 2\langle O_\theta \rangle_\theta) \rangle_\theta, \quad (\text{A.11})$$

and the gradient of S_2 is:

$$\nabla_\theta S_2 = -\frac{\langle R_\theta \Delta O_\theta \rangle_\theta + \langle R_\theta (O_\theta(x) + O_\theta(y) - 2\langle O_\theta \rangle_\theta) \rangle_\theta}{\langle R \rangle_\theta}. \quad (\text{A.12})$$

Effective loss function for automatic differentiation

Expression (A.12) contains Jacobians O_θ which in practice are expensive to compute explicitly for neural networks with many parameters. Instead, we construct an effective loss function

$\mathcal{L}_{S_2}(\theta)$ such that $\nabla_{\theta}\mathcal{L}_{S_2} = \nabla_{\theta}S_2$, so that automatic differentiation reproduces the correct gradient without needing to compute the Jacobians explicitly.

Defining the normalized weights $w_i = R_i/\langle R \rangle_{\theta}$, treated as constants during differentiation (using `stop_gradient`), the effective loss function decomposes into two terms:

$$\mathcal{L}_{S_2}(\theta) = \mathcal{L}_1(\theta) + \mathcal{L}_2(\theta), \quad (\text{A.13})$$

with

$$\mathcal{L}_1(\theta) = -\frac{2}{n} \sum_i w_i \operatorname{Re}[\ln \Psi_{\theta}(x'_i) + \ln \Psi_{\theta}(y'_i)], \quad (\text{A.14})$$

$$\mathcal{L}_2(\theta) = +\frac{2}{n} \sum_i \operatorname{Re}[\ln \Psi_{\theta}(x_i) + \ln \Psi_{\theta}(y_i)]. \quad (\text{A.15})$$

We demonstrate that $\nabla_{\theta}\mathcal{L}_{S_2} = \nabla_{\theta}S_2$.

Term $\nabla_{\theta}\mathcal{L}_1$. With w_i treated as constant:

$$\nabla_{\theta}\mathcal{L}_1 = -\frac{2}{n} \sum_i w_i \operatorname{Re}[O_{\theta}(x'_i) + O_{\theta}(y'_i)] = -\frac{\langle R_{\theta} (O_{\theta}(x') + O_{\theta}(y')) \rangle_{\theta}}{\langle R_{\theta} \rangle_{\theta}}. \quad (\text{A.16})$$

Term $\nabla_{\theta}\mathcal{L}_2$.

$$\nabla_{\theta}\mathcal{L}_2 = \frac{2}{n} \sum_i \operatorname{Re}[O_{\theta}(x_i) + O_{\theta}(y_i)] = 2\langle O_{\theta}(x) + O_{\theta}(y) \rangle_{\theta}. \quad (\text{A.17})$$

Sum of both terms.

$$\begin{aligned} \nabla_{\theta}\mathcal{L}_{S_2} &= -\frac{\langle R_{\theta} (O_{\theta}(x') + O_{\theta}(y')) \rangle_{\theta}}{\langle R_{\theta} \rangle_{\theta}} + 2\langle O_{\theta}(x) + O_{\theta}(y) \rangle_{\theta} \\ &= -\frac{\langle R_{\theta} \Delta O_{\theta} \rangle_{\theta}}{\langle R_{\theta} \rangle_{\theta}} - \frac{\langle R_{\theta} (O_{\theta}(x) + O_{\theta}(y)) \rangle_{\theta}}{\langle R_{\theta} \rangle_{\theta}} + 2\langle O_{\theta}(x) + O_{\theta}(y) \rangle_{\theta} \\ &= -\frac{\langle R_{\theta} \Delta O_{\theta} \rangle_{\theta}}{\langle R_{\theta} \rangle_{\theta}} - \frac{\langle R_{\theta} (O_{\theta}(x) + O_{\theta}(y)) \rangle_{\theta} - \langle R_{\theta} \rangle_{\theta} \cdot 2\langle O_{\theta}(x) + O_{\theta}(y) \rangle_{\theta}}{\langle R_{\theta} \rangle_{\theta}}, \end{aligned} \quad (\text{A.18})$$

where in the last step we used that $2\langle O_{\theta} \rangle_{\theta} = \langle R_{\theta} \rangle_{\theta} \cdot 2\langle O_{\theta} \rangle_{\theta} / \langle R_{\theta} \rangle_{\theta}$. Recognizing that $\langle R_{\theta}(O_{\theta} - 2\langle O_{\theta} \rangle_{\theta}) \rangle_{\theta} = \langle R_{\theta}(O_{\theta} - \langle O_{\theta} \rangle_{\theta}) \rangle_{\theta} - \langle R_{\theta} \rangle_{\theta} \langle O_{\theta} \rangle_{\theta}$ and grouping:

$$\nabla_{\theta}\mathcal{L}_{S_2} = -\frac{\langle R_{\theta} \Delta O_{\theta} \rangle_{\theta} + \langle R_{\theta}(O_{\theta}(x) + O_{\theta}(y) - 2\langle O_{\theta} \rangle_{\theta}) \rangle_{\theta}}{\langle R_{\theta} \rangle_{\theta}} = \nabla_{\theta}S_2, \quad (\text{A.19})$$

which proves that $\mathcal{L}_{S_2}$ exactly reproduces the gradient (A.12).

A.3 Implementation of the S_2 estimator

The numerical implementation of the estimator (2.26) and its gradient (A.12) is carried out in JAX using automatic differentiation over the effective loss function (A.13). For large systems, the computation is split into fixed-size chunks to control GPU memory usage, combining `jax.lax.map` with `jax.checkpoint` for graph rematerialization:

```
1  @partial(jax.jit, static_argnames=("apply_fun", "chunk_size"))
2  def _renyi2_loss_and_grad(apply_fun, params, model_state,
3                           samples1, samples2, swapped1, swapped2,
4                           chunk_size=256):
5
6      def log_psi(p, s):
7          return apply_fun({"params": p, **model_state}, s)
8
9      n = samples1.shape[0]
10     n_chunks = n // chunk_size
11
12     def reshape(x):
13         return x.reshape(n_chunks, chunk_size, *x.shape[1:])
14
15     s1_c, s2_c, sw1_c, sw2_c = map(
16         reshape, (samples1, samples2, swapped1, swapped2)
17     )
18
19     # Forward pass: log_R without backprop
20     def forward_chunk(bs1, bs2, bsw1, bsw2):
21         def single(s1, s2, sw1, sw2):
22             lo1 = log_psi(params, s1[None])[0]
23             lo2 = log_psi(params, s2[None])[0]
24             ls1 = log_psi(params, sw1[None])[0]
25             ls2 = log_psi(params, sw2[None])[0]
26             return jnp.real(ls1 + ls2 - lo1 - lo2)
27         return jax.vmap(single)(bs1, bs2, bsw1, bsw2)
28
29     log_R = jax.lax.map(
30         lambda x: forward_chunk(*x), (s1_c, s2_c, sw1_c, sw2_c)
31     ).reshape(n)
32     log_R = jax.lax.stop_gradient(log_R)
```

```

33     R_mean = jax.lax.stop_gradient(jnp.mean(jnp.exp(log_R)))
34     S2 = -jnp.log(jnp.abs(R_mean))
35
36     # Backward pass: chunked gradient with rematerialization
37     def loss_fn(p):
38         log_R_c = log_R.reshape(n_chunks, chunk_size)
39
40         @jax.checkpoint
41         def grad_chunk(bs1, bs2, bsw1, bsw2, blog_R):
42             def single(s1, s2, sw1, sw2, log_Ri):
43                 lo1 = log_psi(p, s1[None])[0]
44                 lo2 = log_psi(p, s2[None])[0]
45                 ls1 = log_psi(p, sw1[None])[0]
46                 ls2 = log_psi(p, sw2[None])[0]
47                 w_i = jax.lax.stop_gradient(
48                     jnp.exp(log_Ri) / R_mean
49                 )
50                 return (
51                     -2.0 * w_i * jnp.real(ls1 + ls2)
52                     + 2.0 * jnp.real(lo1 + lo2)
53                 )
54             return jax.vmap(single)(bs1, bs2, bsw1, bsw2, blog_R)
55
56         contributions = jax.lax.map(
57             lambda x: grad_chunk(*x),
58             (s1_c, s2_c, sw1_c, sw2_c, log_R_c)
59         )
60         return jnp.mean(contributions)
61
62     grad_S2 = jax.grad(loss_fn)(params)
63     return S2, grad_S2

```

Listing A.1: Chunked estimator of the S_2 gradient.

The forward pass computes $\log R_i$ with `stop_gradient` to avoid materializing the full graph. The backward pass recomputes the amplitudes within each chunk using `jax.checkpoint`, which rematerializes activations during the backward pass instead of storing them, reducing memory usage from $\mathcal{O}(n)$ to $\mathcal{O}(\text{chunk_size})$.

B. NQS training implementation

This appendix collects the most relevant code fragments of the implementation. The complete training is orchestrated by the function `free_energy_minimize` (Listing B.1), which delegates the joint calculation of F_2 and $\nabla_{\theta} F_2$ to the observable `FreeRenyiEnergyObservable` (Listing B.2).

Main optimization loop

```
1  for step in range(n_steps):
2      # Gradient of F_2 = <H> - T*S_2
3      F_stats, F_grad = vstate.expect_and_grad(free_renyi_op)
4
5      # Optimizer step (sign-SGD with linear lr)
6      updates, opt_state = gradient_transform.update(
7          F_grad, opt_state, vstate.parameters
8      )
9      vstate.parameters = optax.apply_updates(
10         vstate.parameters, updates
11     )
12
13     F_val = float(F_stats.mean.real)
14     free_energy_history.append(F_val)
15
16     # Save the best visited state
17     if F_val < best_F:
18         best_F = F_val
19         best_params = vstate.parameters
20
21     # Restore best parameters and re-evaluate with high statistics
22     vstate.parameters = best_params
23     E_best = float(vstate.expect(Hamiltonian).mean.real)
24     S2_best = renyi2_entropy_sampled(
25         vstate, partition, n_samples_full, chunk_size=chunk_size
26     )
27     best_F = E_best - T * S2_best
```

Listing B.1: Optimization loop for the Rényi-2 free energy.

The observable `free_renyi_op` is an instance of `FreeRenyiEnergyObservable`, which implements the NetKet interface so that `vstate.expect_and_grad` can simultaneously compute F_2 and $\nabla_\theta F_2$ reusing a single set of samples. At the end of training, the best found parameters are restored and $\langle \hat{H}_{SA} \rangle$ and S_2 are re-evaluated with independent high-statistics estimates.

Rényi-2 free energy observable

It is recommended to read Appendix A beforehand since it explains how S_2 and its gradient are computed. Directly computing the gradient of $F_2 = \langle \hat{H}_{SA} \rangle - TS_2$ requires evaluating $\log |\Psi(\theta, \sigma)|$ for $2N_{\text{samples}}$ configurations simultaneously, which easily exceeds available GPU memory. To avoid this, the observable splits the batch into chunks and applies materialization (`jax.checkpoint`) on the backward pass, so that intermediate activations are recomputed rather than stored.

```

1  class FreeRenyiEnergyObservable(AbstractObservable):
2      """
3      Observable F = E - T*S_2 whose gradient is computed in a fused
4      manner
5      in a single jit, reusing the vstate samples.
6      """
7      def __init__(self, hilbert, H, partition, T, chunk_size=128):
8          super().__init__(hilbert)
9          self.H = H; self.partition = partition
10         self.T = T; self.chunk_size = chunk_size
11
12     @nk.vqs.expect_and_grad.dispatch
13     def expect_and_grad_free_renyi(vstate, op, chunk_size, **kwargs):
14         # --- Sample preparation ---
15         subsystem_sites = jnp.array(op.partition, dtype=int)
16         all_sites = jnp.arange(vstate.hilbert.size)
17         complement_sites = jnp.setdiff1d(all_sites, subsystem_sites)
18
19         all_samples = vstate.samples.reshape(-1, vstate.hilbert.size)
20         n = all_samples.shape[0] // 2
21         samples1, samples2 = all_samples[:n], all_samples[n:]

```



```

22
23 # Swap subsystem A between the two copies
24 swapped1 = jnp.concatenate(
25     [samples2[:, subsystem_sites], samples1[:,
26         complement_sites]], axis=1
27 )
28 swapped2 = jnp.concatenate(
29     [samples1[:, subsystem_sites], samples2[:,
30         complement_sites]], axis=1
31 )
32
33 # --- Chunked E_loc ---
34 cs, n_chunks = op.chunk_size, n // op.chunk_size
35 E_loc = jnp.concatenate([
36     _compute_E_loc_jit(
37         vstate._apply_fun, vstate.parameters, vstate.
38         model_state,
39         samples1[i*cs:(i+1)*cs],
40         *op.H.get_conn_padded(samples1[i*cs:(i+1)*cs]),
41         chunk_size=cs,
42     )
43     for i in range(n_chunks)
44 ])
45 E_loc_sg = jax.lax.stop_gradient(E_loc)
46
47 # --- Fused gradient of F_2 ---
48 E_mean, S2, grad_F = _free_renyi_grad_jit(
49     vstate._apply_fun, vstate.parameters, vstate.model_state,
50     samples1, samples2, E_loc_sg,
51     swapped1, swapped2, op.T, op.chunk_size,
52 )
53
54 F = float(E_mean) - op.T * float(S2)
55 return nk.stats.statistics(jnp.array([[F]])), grad_F

```

Listing B.2: FreeRenyiEnergyObservable: fused computation of F_2 and $\nabla_\theta F_2$.

Gradient consistency validation

At the end of training, the reliability of the stochastic estimator is estimated by computing the cosine similarity between $N_{\text{rep}} = 10$ independent estimates of $\nabla_{\theta} S_2$:

$$\cos(\nabla^{(i)}, \nabla^{(j)}) = \frac{\nabla^{(i)} \cdot \nabla^{(j)}}{\|\nabla^{(i)}\| \|\nabla^{(j)}\|}. \quad (\text{B.1})$$

A value close to 1 confirms that the stochastic variance is small compared to the gradient magnitude, which validates the choice of N_{samples} .

C. MRE transition temperatures

For $\alpha = 2$, the analytical solution of the MRE has the form

$$\rho_2 = \frac{1}{Z_2} \sum_{k: E_k < E_\perp} (E_\perp - E_k) |E_k\rangle \langle E_k|, \quad (\text{C.1})$$

where the cutoff $E_\perp$ is determined self-consistently from the condition

$$E_\perp = \frac{2}{\beta} + \bar{E} = 2T + \bar{E}, \quad (\text{C.2})$$

with $\bar{E} = \langle H \rangle = \text{Tr}(\rho_2 H)$. We denote by $E_0 < E_1 < E_2 < \dots$ the eigenvalues of the Hamiltonian ordered from lowest to highest (collapsing degenerate levels).

Temperature of the n -th jump

Suppose that for a given temperature T the cutoff satisfies $E_{n-1} < E_\perp < E_n$, so that only the first n levels are active. The mean energy is then

$$\bar{E}^{(n)}(E_\perp) = \frac{\sum_{k=0}^{n-1} E_k (E_\perp - E_k)}{\sum_{k=0}^{n-1} (E_\perp - E_k)}. \quad (\text{C.3})$$

The n -th jump occurs at the temperature T_n at which the cutoff exactly reaches E_n , i.e., when $E_\perp = E_n$. Substituting into (C.2):

$$E_n = 2T_n + \bar{E}^{(n)}(E_n). \quad (\text{C.4})$$

Solving for T_n :

$$T_n = \frac{E_n - \bar{E}^{(n)}(E_n)}{2} = \frac{1}{2} \left(E_n - \frac{\sum_{k=0}^{n-1} E_k (E_n - E_k)}{\sum_{k=0}^{n-1} (E_n - E_k)} \right). \quad (\text{C.5})$$

Particular cases

For $n = 1$, only the ground state is active before the jump, so $\bar{E}^{(1)} = E_0$ trivially, and

$$T_1 = \frac{E_1 - E_0}{2}. \quad (\text{C.6})$$

This result can also be obtained directly: near T_1 , the state is a mixture $\rho = (1-p)|E_0\rangle\langle E_0| + p|E_1\rangle\langle E_1|$ with $p \ll 1$. Expanding the Rényi free energy to first order in p ,

$$F_2(p) \approx E_0 + p(E_1 - E_0) - 2Tp + \mathcal{O}(p^2), \quad (\text{C.7})$$

and requiring that the pure state $p = 0$ ceases to be stable, $\partial F_2/\partial p|_{p=0} = 0$, recovers $T_1 = (E_1 - E_0)/2$.

Nature of the discontinuity

At each T_n the level E_n enters the ensemble with zero weight but with non-zero derivative with respect to T . Local observables $\langle O \rangle(T)$ are therefore continuous at T_n , but their derivative $d\langle O \rangle/dT$ exhibits a finite discontinuity, analogous to a second-order phase transition. In particular, the mean energy $\langle H \rangle(T)$ and the entropy $S_2(T)$ are continuous, while the specific heat $C = d\langle H \rangle/dT$ has a finite jump at each T_n .

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